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Semester	4th
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Subject	Computer Networks
Course Code	CS-2203
Credit Hours	3+1
Lab	06
Date	24-Apr-2026
Department of Software Engineering	

LAB MANUAL – COMPUTER NETWORKS LAB 4

1. TITLE

Basic Router Access and CLI Configuration Using Cisco Packet Tracer

2. Objective

- Access router using CLI.
- Understand Cisco IOS modes.
- Configure passwords.
- Enable password encryption.
- View configuration files.
- Save router configuration.

3. Software Required

- Cisco Packet Tracer
- PC / Laptop

4. Apparatus Required

- 1 Router
- Computer System
- Console cable (real lab)

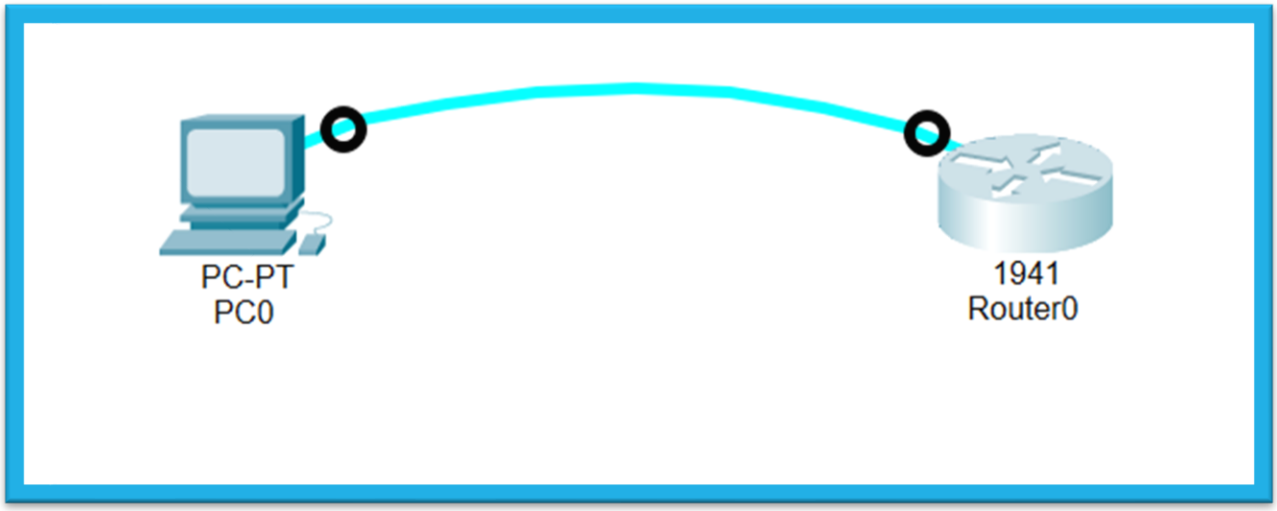
5. Theory

Routers must be configured before use. Configuration is done through **CLI (Command Line Interface)**. Cisco IOS has different modes:

Mode	Prompt	Purpose
User EXEC	Router>	Basic commands
Privileged EXEC	Router#	Advanced commands
Global Config	Router(config)#	Configuration settings

6. Procedure (Lab Task)

Step 1: Open Cisco Packet Tracer



- Place one router in workspace.
- Click router.
- Open **CLI** tab.
- Press Enter.

Output:

Router>

Step 2: Enter Privileged Mode

enable

Output:

```
         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router> enable
Router#
```

Step 3: Enter Global Configuration Mode

configure terminal

Output:

```
Router# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
```

Step 4: Configure Enable Password

enable password cisco123

```
Router(config)# enable password cisco123
Router(config)#
```

Step 5: Enable Password Encryption

service password-encryption

```
Router(config)# enable password cisco123
Router(config)# service password-encryption
Router(config)# enable secret class456
```

Step 6: Configure Enable Secret Password

enable secret admin123

```
Router(config)# enable password cisco123
Router(config)# service password-encryption
Router(config)# enable secret class456
```

Step 7: Exit Configuration Mode

end

```
Router(config)#end
```

Step 8: Display Running Configuration

show running-config

```
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show running-config
Building configuration...

Current configuration : 693 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$fUPSchZrwvVDQvkO1sLfl/
enable password 7 0822455D0A16544541
!
!
!
!
!
!
ip cef
 --More--
```

Step 9: Display Startup Configuration

show startup-config

```
Router# show startup-config
startup-config is not present
Router#
```

Step 10: Save Configuration

copy running-config startup-config

Press Enter.

Output:

Building configuration...

[OK]

```
Router# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

```
Router# show startup-config
startup-config is not present
Router# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router# show startup-config
Using 693 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$fUPSchZrwvVDQvkO1sLfl/
enable password 7 0822455D0A16544541
!
!
!
!
!
!
ip cef
no ipv6 cef
!
```

7. Commands Used

enable

configure terminal

enable password cisco123

service password-encryption

enable secret admin123

end

show running-config

show startup-config

copy running-config startup-config

9. RESULT

Router configuration was completed successfully using Cisco CLI commands.